



Moriah College
MATHEMATICS DEPARTMENT

Year 12

Mathematics Pre-Trial 2007

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Board-approved calculators may be used
- A table of standard integrals is provided on page and a detachable sheet is provided on page.
- All necessary working should be shown in every question

Total marks (120)

- Attempt Questions 1–10
- All questions are of equal value
- Use a **SEPARATE** answer sheet for each question

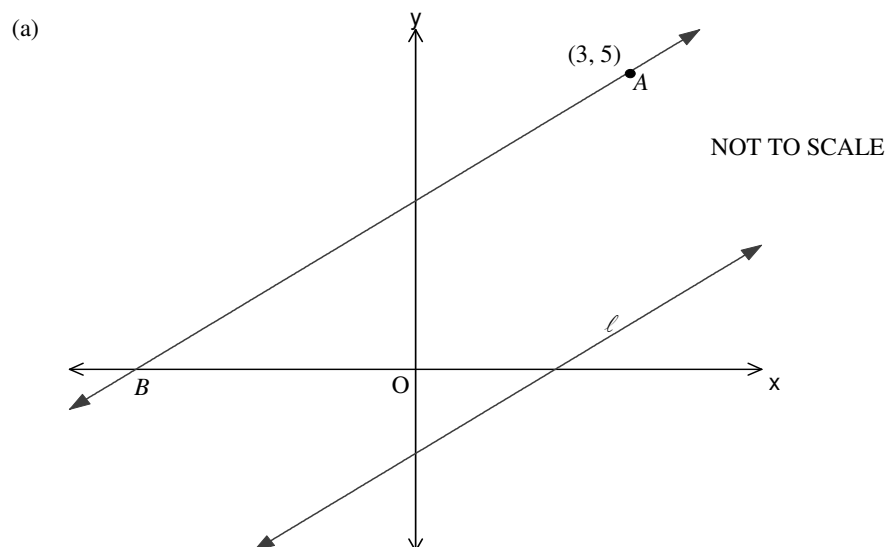
STUDENT NUMBER: _____ **CLASS TEACHER:** _____

Question 1 (12 marks). Use a SEPARATE Writing Booklet.

	Marks
(a) Evaluate e^{-3} correct to three significant figures.	2
(b) Solve $ 3x + 2 \leq 5$	2
(c) Fully factorise $27x^3 - 64y^3$	2
(d) Differentiate with respect to x : $y = \frac{x^2}{2} - \frac{4}{x^2}$	2
(e) Find the fractional equivalent of $0.\dot{6}8$.	2
(f) Find integers a and b such that: $(3 - \sqrt{2})^2 = a - b\sqrt{2}$	2

Question 2 (12 marks). Use a SEPARATE Writing Booklet.

Marks

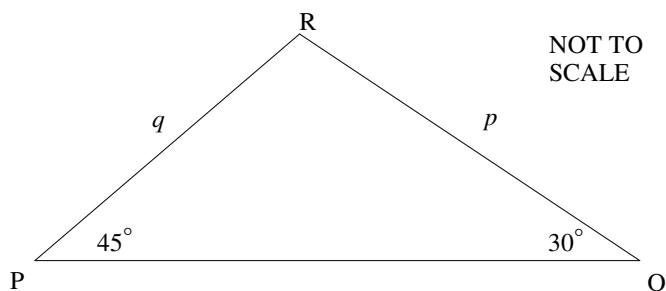


- (a) The diagram shows the points $A(3,5)$ and B , where the line AB cuts the x -axis at B . The line ℓ has equation $3x - 5y - 8 = 0$ and is parallel to AB .
- Find the gradient of the line AB . 1
 - Find the equation of the line AB . 1
 - Find the coordinates of point B . 1
 - Write down the size of $\angle ABO$ correct to the nearest degree. 1
 - Another point $C(-2, 2)$ also lies on AB . Find the exact length of the interval AC . 1
 - Find the exact perpendicular distance of C from the line ℓ . 2
 - Find the exact area of the triangle formed by the points A , C and any point P , on the line ℓ . 1
 - Explain why the area of triangle ACP is constant, regardless of the position of P on the line ℓ . 1
- (b) Find the equation of the normal to the curve $y = 2x^2 - 5x + 1$ at the point $(2, -1)$. 3

Question 3 (12 marks). Use a SEPARATE Writing Booklet.

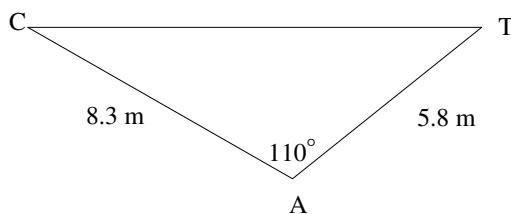
Marks
3

- (a) In $\triangle PQR$ below, $\angle RPQ = 45^\circ$ and $\angle RQP = 30^\circ$.



Find the exact value of $\frac{p}{q}$.

- (b) In $\triangle CAT$ below, $CA = 8.3$ m, $AT = 5.8$ m and $\angle TAC = 110^\circ$.



- (i) Find the length of CT correct to one decimal place.

2

- (ii) Find the size of the smallest angle correct to the nearest degree.

2

- (c) (i) Prove that $2\cos^2 B + 3\sin^2 B - 2 = \sin^2 B$ is true for all values of B.

3

- (ii) Hence or otherwise, solve: $2\cos^2 A + 3\sin^2 A - 3 = 0$ for $0^\circ \leq A \leq 360^\circ$.

2

Question 4 (12 marks). Use a SEPARATE Writing Booklet.

	Marks
(a) Find: $\lim_{x \rightarrow 2} \frac{2x^2 - 3x - 2}{x - 2}$	2
(b) Differentiate:	
i. $y = 5\sqrt{x} - \frac{3}{x^4}$	2
ii. $f(x) = \frac{2x + 3}{3x + 2}$	2
(c) A function $y = f(x)$ has $\frac{d^2y}{dx^2} = 6x - 2$ and a stationary point at $(1, 2)$. Find the equation of $f(x)$.	3
(d) Solve for x : $4^x - 9 \times 2^x + 8 = 0$	3

Question 5 (12 marks). Use a SEPARATE Writing Booklet.

	Marks
(a) The sum of the first three terms of a geometric series is 19 and the sum to infinity is 27.	
Find:	
i. the common ratio.	2
ii. the first term.	1
iii. the fifth term.	1
(b) The first four terms of a series are 3, x , y and 192.	
Find the values of x and y if the series is:	
i. Arithmetic	2
ii. Geometric	2
(c) The sum of n terms of a certain series is given by:	
$S_n = 2n(n + 1).$	
i. Find the first three terms of this sequence.	2
ii. Which term of the sequence is 124?	2

Question 6 (12 marks). Use a SEPARATE Writing Booklet.

Marks

(a) Consider the parabola $y = -4x^2 - 16x - 15$,

i. Find the coordinates of its vertex.

2

ii. Find the coordinates of its focus.

2

(b) α and β are the roots of the equation $2x^2 + 5x - 5 = 0$. Find the values of:

i. $\alpha + \beta$

1

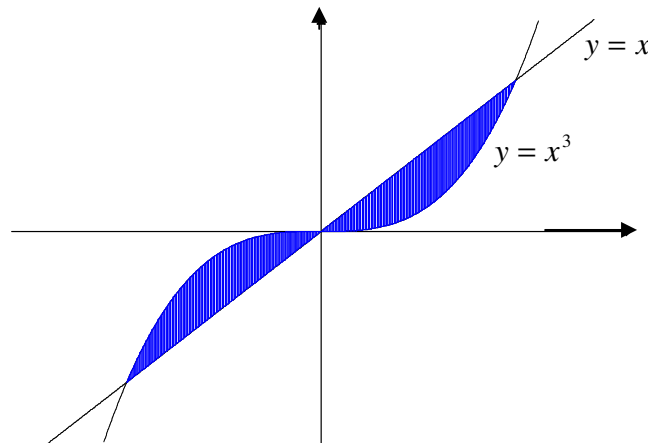
ii. $\alpha\beta$

1

iii. $\alpha^{-1} + \beta^{-1}$

2

(c)



i. Show that $y = x^3$ is an odd function.

1

ii. Hence, or otherwise, find the area between the curves $y = x$ and $y = x^3$.

3

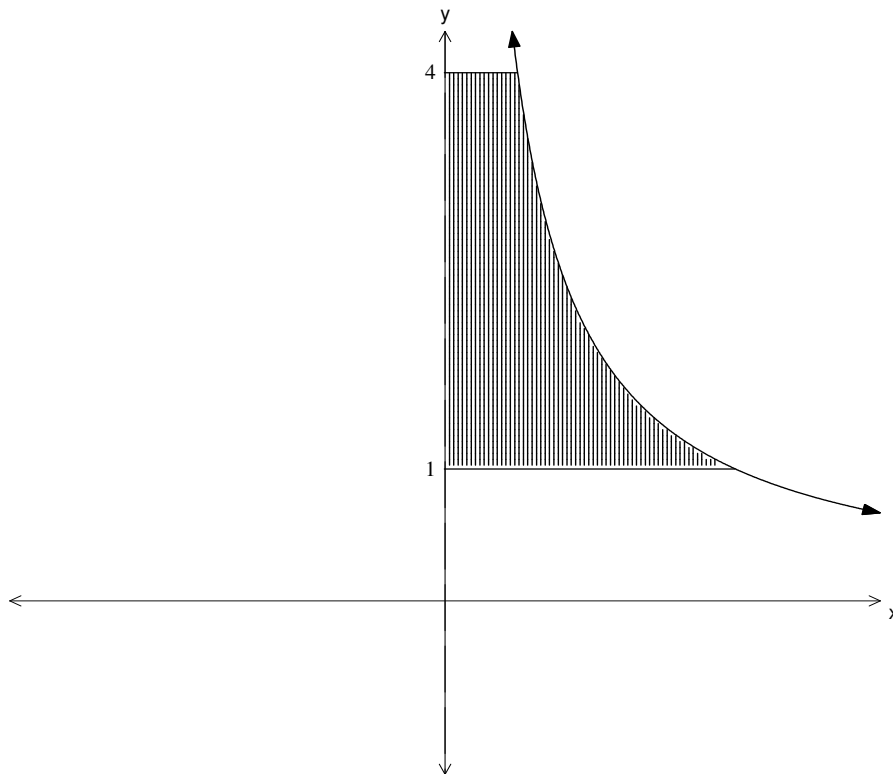
Question 7 (12 marks). Use a SEPARATE Writing Booklet.

	Marks
(a) Consider the curve given by $y = 3x^2 + x^3 - 9x - 5$.	
i. Find $\frac{dy}{dx}$.	1
ii. Find the coordinates of the two stationary points.	2
iii. Determine the nature of the stationary points.	2
iv. Sketch the graph of the function for the domain $-5 \leq x \leq 3$.	2
(b) Evaluate the following integral, giving your answer in exact form:	2
$\int_0^1 e^{2-3x} dx$	
(c) Differentiate $y = e^{-x^3}$ and hence or otherwise find $\int_1^2 x^2 e^{-x^3} dx$ correct to three decimal places.	3

Question 8 (12 marks). Use a SEPARATE Writing Booklet.

Marks

- (a) A point P moves so that the distance from the point $A(-6, -2)$, is three times the distance from the point $B(2, 2)$.
- i. Find the equation that describes the locus of the point P . **4**
- ii. Describe this locus geometrically. **3**
- (b) The area enclosed by the curve $xy = 3$, the y -axis and the lines $y = 1$ and $y = 4$, is rotated about the y -axis.



Find the volume of the solid of revolution correct to three decimal places.

5

Question 9 (12 marks). Use a SEPARATE Writing Booklet.

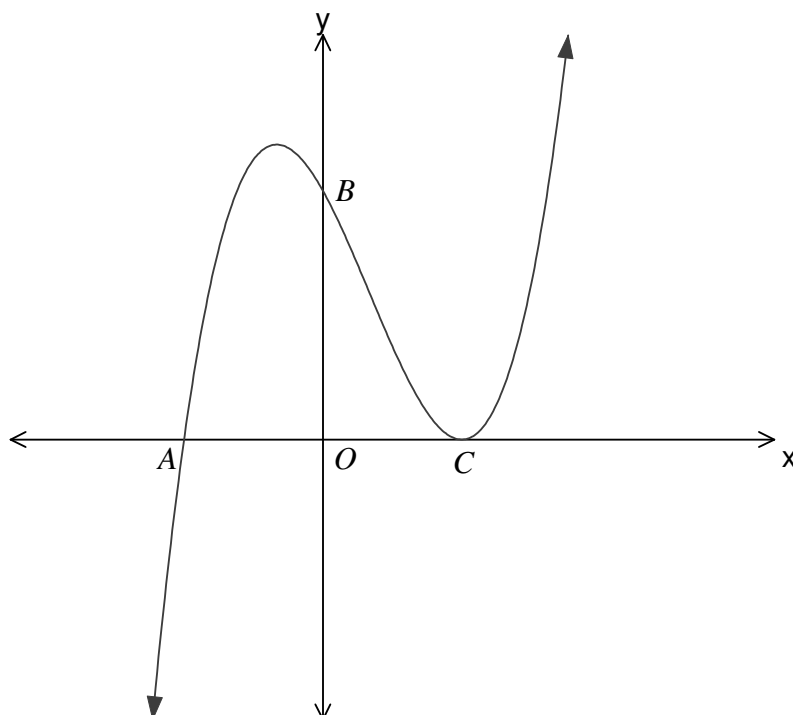
Marks

(a) Differentiate the following functions:

i. $y = (e^{4x+3} - 3)^4$ **2**

ii. $y = \frac{x}{e^x}$ **2**

(b) The diagram shows the sketch of $y = (x+1)(x-1)^2$.



i. Find the coordinates of the points A, B and C, the intercepts on the axes. **2**

ii. Show that OB divides the area between the curve and the line AC in the ratio of 11:5. **3**

(c) The following table gives values of $f(x) = x \log_e x$.

x	1	2	3	4	5
$f(x)$	0	1.39	3.30	5.55	8.05

Use Simpson's Rule and these five functional values in the table above to find

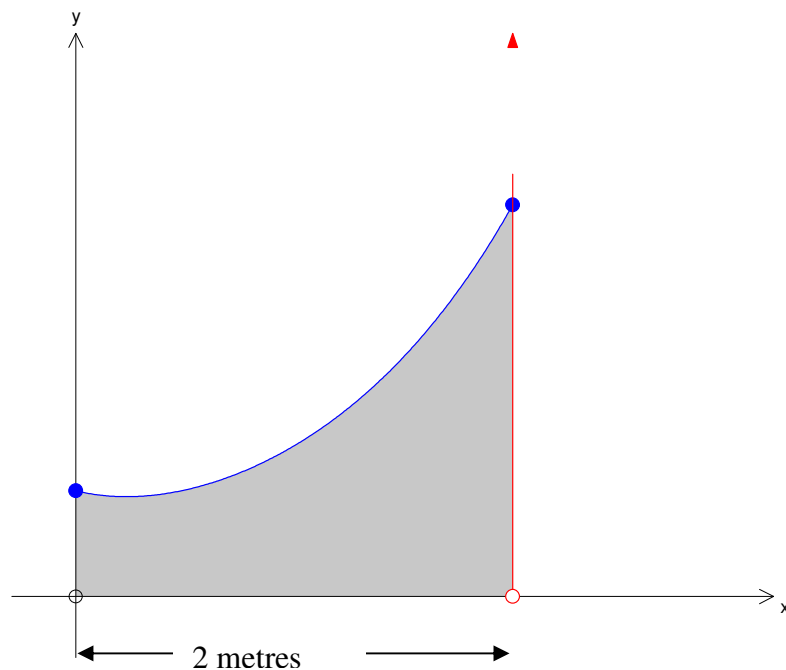
an approximation of $\int_1^5 x \log_e x \, dx$ correct to two decimal places. **3**

Question 10 (12 marks). Use a SEPARATE Writing Booklet.

Marks

- (a) Consider the function $f(x) = \frac{1}{2}(e^x + e^{-x})$
- Show that the curve represents an even function. 1
 - Show that the function has only one stationary point and determine its nature. 3
 - Show that the function has no points of inflexion. 1
 - Hence sketch the curve. 1

- (b) A metal chain hangs between two walls, which stand two metres apart, as shown in the diagram below. The height of this chain above the ground, is given by the equation: $y = e^{-2x} + e^x$ (x and y are measured in metres)



Answer the following questions, giving all answers correct to 2 decimal places:

- How far from the ground is the chain hooked to the right wall? 2
- Find the area of the shaded region. 4

END OF PAPER

MATHEMATICS PRE-TRIAL 2007

Solutions

Question 1

(a) $e^{-3} = 0.0498$ (3 significant figures)

(b)

$$\begin{aligned} -5 &\leq 3x + 2 \leq 5 \\ -7 &\leq 3x \leq 3 \\ \frac{-7}{3} &\leq x \leq 1 \end{aligned}$$

(c)

$$\begin{aligned} (3x)^3 - (4y)^3 &\Rightarrow \\ (3x - 4y)(9x^2 + 12xy + 16y^2) \end{aligned}$$

(d)

$$\begin{aligned} y &= \frac{1}{2}x^2 - 4x^{-2} \Rightarrow \\ y' &= x + 8x^{-3} \\ y' &= x + \frac{8}{x^3} \end{aligned}$$

(e)

$$\begin{aligned} \text{Let: } x &= 0.\dot{6}\dot{8} \\ \text{So: } 10x &= 6.\dot{8} & [1] \\ \text{And: } 100x &= 68.\dot{8} & [2] \end{aligned}$$

Subtracting [1] from [2] yields:

$$\begin{aligned} 90x &= 62 \\ x &= \frac{62}{90} = \frac{31}{45} \end{aligned}$$

(f)

$$\begin{aligned} (3 - \sqrt{2})^2 &= a - b\sqrt{2} \\ 9 - 6\sqrt{2} + 2 &= a - b\sqrt{2} \\ 11 - 6\sqrt{2} &= a - b\sqrt{2} \end{aligned}$$

Comparing the rational coefficients
and the irrational coefficients
yields:

$$\begin{aligned} a &= 11 \\ b &= 6 \end{aligned}$$

Question 2

(a)

i. Rearrange: $3x - 5y - 8 = 0$

$$y = \frac{3}{5}x - \frac{8}{5} \Rightarrow$$

$$m = \frac{3}{5}$$

$$\begin{aligned} y &= \frac{3}{5}x + b \\ 5 &= \frac{3}{5} \times 3 + b \\ \Rightarrow 25 &= 9 + 5b \\ b &= \frac{16}{5} \end{aligned}$$

$$\begin{aligned} \text{ii. } \frac{y-5}{x-3} &= \frac{3}{5} \\ 5y - 3x - 16 &= 0 \end{aligned}$$

iii. $y = \frac{3}{5}x + \frac{16}{5}$
Substitute $y=0$ in the equation of AB:

$$-3x - 16 = 0$$

$$x = -\frac{16}{3}$$

$$B: \left(-\frac{16}{3}, 0 \right) \quad \left(-5\frac{1}{3}, 0 \right)$$

iv. Let $\angle ABO = \alpha$, then:

$$\tan \alpha = \frac{3}{5} \Rightarrow \alpha = 31^\circ \text{ (nearest deg)}$$

v. A: (3,5) and C: (-2,2), so the length AB is:

$$AC = \sqrt{(3+2)^2 + (5-2)^2}$$

$$AC = \sqrt{5^2 + 3^2}$$

$$AC = \sqrt{34} \text{ units}$$

$$d = \frac{|ax+by+c|}{\sqrt{a^2+b^2}}$$

vi. The distance of line: $3x - 5y - 8 = 0$ from the

$$\begin{aligned} &\frac{|3(-2) + (-5)(2) - 8|}{\sqrt{3^2 + (-5)^2}} \\ &= \frac{24}{\sqrt{34}} = \frac{12\sqrt{34}}{17} \end{aligned}$$

vii. Area:

$$= \frac{1}{2} \times \sqrt{34} \times \frac{24}{\sqrt{34}} = 12 \text{ u}^2$$

viii. The lines are parallel and, therefore, the distance between the two lines is constant.

$$(b) \quad y' = 4x - 5 \quad \text{at } x=2: y' = 3$$

$$\therefore m_1 = -\frac{1}{3}$$

$$\therefore y = -\frac{1}{3}x + c$$

$$(2, -1): -1 = -\frac{2}{3} + c$$

$$c = -\frac{1}{3}$$

$$\therefore y = -\frac{1}{3}x - \frac{1}{3}$$

$$\textcircled{OR} \quad 3y + x + 1 = 0. \quad \boxed{3}$$

Question 3

- a) Applying the sine rule to the triangle RPQ :

$$\frac{p}{\sin 45^\circ} = \frac{q}{\sin 30^\circ} \quad |$$

$$\frac{p}{q} = \frac{\sin 45^\circ}{\sin 30^\circ} = \frac{\frac{\sqrt{2}}{2}}{\frac{1}{2}} = \sqrt{2} \quad \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2} \times \sqrt{2}} \quad |$$

b)

- i. Applying the cosine rule to the triangle ACT :

$$CT^2 = 8.3^2 + 5.8^2 - 2 \times 8.3 \times 5.8 \times \cos 110^\circ \quad |$$

$$CT = 11.6m \quad (1dp) - \frac{1}{2} \quad |$$

- ii. The smallest angle lies opposite the smallest side, so angle TCA is the smallest angle.

Let $\angle TCA = \theta$ Applying the sine rule to the triangle TCA :

$$\frac{5.8}{\sin \theta} = \frac{11.6}{\sin 110^\circ} \Rightarrow$$

$$\frac{5.8 \times \sin 110^\circ}{11.6} = \sin \theta$$

$$0.468 = \sin \theta$$

$$\theta = 28^\circ \quad (\text{nearest deg})^{-\frac{1}{2}} \quad |$$

$$\frac{\sin \theta}{5.8} = \frac{\sin 110^\circ}{11.6} \quad |$$

$$\sin \theta = \frac{5.8 \sin 110^\circ}{11.6}$$

use the identity:

$$\cos^2 B + \sin^2 B = 1$$

c) (i) LHS:

$$= 2 \cos^2 B + 3 \sin^2 B - 2$$

$$= 2(1 - \sin^2 B) + 3 \sin^2 B - 2 \quad 2$$

$$= 2 - 2 \sin^2 B + 3 \sin^2 B - 2 \quad |$$

$$= \sin^2 B$$

So, LHS = RHS

(ii)

$$2 \cos^2 A + 3 \sin^2 A - 2 - 1 = 0 \quad |$$

$$\sin^2 A - 1 = 0$$

$$\sin^2 A = 1$$

$$\sin A = \pm 1$$

$$\sin A = 1 \Rightarrow A = 90^\circ$$

$$\sin A = -1 \Rightarrow A = 270^\circ \quad \} \quad |$$

If no $\pm$, then $\square$ 1 mark

5

4

Question 4

$$\begin{aligned} \text{(a)} \quad & \lim_{x \rightarrow 2} \frac{2x^2 - 3x - 2}{x - 2} \\ &= \lim_{x \rightarrow 2} \frac{(x-2)(2x+1)}{x-2} \\ &= \lim_{x \rightarrow 2} (2x+1) = 5 \end{aligned}$$

(b)

Rewrite: $y = 5x^{\frac{1}{2}} - 3x^{-4}$

$$y' = \frac{5}{2}x^{-\frac{1}{2}} + 12x^{-5}$$

$$y' = \frac{5}{2\sqrt{x}} + \frac{12}{x^5}$$

Let $u = 2x+3$ $v = 3x+2$
 $u' = 2$ $v' = 3$

Use the quotient rule:

$$f'(x) = \frac{u'v - v'u}{(v)^2}$$

$$f'(x) = \frac{2(3x+2) - 3(2x+3)}{(3x+2)^2}$$

$$f'(x) = \frac{-5}{(3x+2)^2}$$

(c) Given $f''(x) = 6x - 2$

$$\Rightarrow f'(x) = 3x^2 - 2x + C$$

Also, given stationary

point: $f'(1) = 0$

$$\Rightarrow 0 = 3 - 2 + C$$

$$\Rightarrow C = -1$$

So, $f'(x) = 3x^2 - 2x - 1$.

Therefore: $f(x) = x^3 - x^2 - x + D$

The point (1,2) lies on the graph so

$f(1) = 2$. Substitute:

$$2 = 1^3 - 1^2 - 1 + D$$

$$\Rightarrow D = 3$$

$$f(x) = x^3 - x^2 - x + 3$$

(d) Let $m = 2^x$, so the equation simplifies to:

$$m^2 - 9m + 8 = 0$$

$$(m-1)(m-8) = 0$$

$$m = 1 \quad \text{or} \quad m = 8$$

$$2^x = 1 \quad \text{or} \quad 2^x = 8$$

$$x = 0 \quad \text{or} \quad x = 3$$

Question 5

(a)

i. Given: $\begin{cases} S_3 = 19 \\ S = 27 \end{cases}$

$$\Rightarrow \begin{cases} \frac{a(1-r^3)}{1-r} = 19 & \text{[I]} \\ \frac{a}{1-r} = 27 & \text{[II]} \end{cases}$$

Dividing [I] by [II] yields:

$$1-r^3 = \frac{19}{27} \text{ and that can be simplified to:}$$

$$r^3 = \frac{8}{27} \Rightarrow r = \frac{2}{3}$$

ii. Put in [II] and find the value of the first term: $a=9$

iii. And so, for a general formula:

$$T_n = 9 \times \left(\frac{2}{3}\right)^{n-1} \Rightarrow$$

$$T_5 = \frac{16}{9}$$

(b) Let the sequence of numbers be:

3, x, y, 192

We know that $\begin{cases} a = 3 \\ T_4 = 192 \end{cases}$

i. If the sequence is an AP then: $192 = 3 + 3d$
 $d = 63$ and then: $x = 66$ and $y = 129$

i. If the sequence is a GP then:

$$192 = 3 \times r^3$$

$$64 = r^3$$

$$4 = r$$

And so, $x = 12$ and $y = 48$

(c)

$$S_1 = T_1 = 2 \times 2$$

$$\therefore T_1 = 4$$

i. $S_2 = T_1 + T_2 \Rightarrow$

$$12 = 4 + T_2$$

$$\therefore T_2 = 8$$

$$S_3 = 24$$

$$S_3 = T_1 + T_2 + T_3 \Rightarrow$$

$$24 = 4 + 8 + T_3$$

$$\therefore T_3 = 12$$

clearly an AP emerging, for which the first term = 4 and the common difference is 4:

$$T_n = 4 + 4(n-1)$$

$$T_n = 4n \Rightarrow$$

$$124 = 4n$$

$$31 = n$$

So, the 31st term of the given sequence is 124

Question 6

(a)

Rewrite the equation of the parabola:

$$y = -4(x^2 + 4x + 4) + 1$$

$$y - 1 = -4(x + 2)^2$$

$$\frac{1}{4}(y - 1) = -(x + 2)^2$$

The vertex is at: $(-2, 1)$

Comparing the parabola to: $4a(y - n) = -(x - p)^2$

$$\text{We get: } 4a = \frac{1}{4} \text{ or: } a = \frac{1}{16}$$

Therefore the focus is at: $\left(-2, \frac{15}{16}\right)$

(b) $\alpha + \beta = \frac{-b}{a} = \frac{-5}{2}$

$$\alpha\beta = \frac{c}{a} = \frac{-5}{2}$$

$$\alpha^{-1} + \beta^{-1} = \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = 1$$

(c)

i. Let: $f(x) = x^3$

$$\text{So, } f(-x) = (-x)^3 = -x^3$$

$$f(-x) = -f(x) \text{ [Hence, odd function]}$$

ii. Find points of intersection between $y = x^3$ and $y = x$

$$x = x^2 \Rightarrow x^3 - x = 0$$

$$\Rightarrow x(x^2 - 1) = 0 \Rightarrow x(x - 1)(x + 1) = 0$$

Intersection at: $(1, 0)$, $(-1, 0)$ and $(0, 0)$

Since the graph is of an odd function, it has a point symmetry and the area shaded in the first quadrant is the same as that in the fourth quadrant. Therefore, it is sufficient to

$$\text{calculate: } A = 2 \int_0^1 (x - x^3) dx$$

$$A = 2 \times \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^1 = 2 \times \frac{1}{4} \text{ so total area:}$$

$$2 \times \frac{1}{4} = \frac{1}{2} \text{ units}^2$$

Question 7

a.

i. $\frac{dy}{dx} = 6x + 3x^2 - 9$ ✓

ii. For stationary points, solve: $\frac{dy}{dx} = 0$ ✓ $\Rightarrow$

$$0 = 6x + 3x^2 - 9$$

$$(-3, 22) \quad \text{or} \quad (1, -10)$$
 ✓

iii. $\frac{d^2y}{dx^2} = 6 + 6x$ ✓

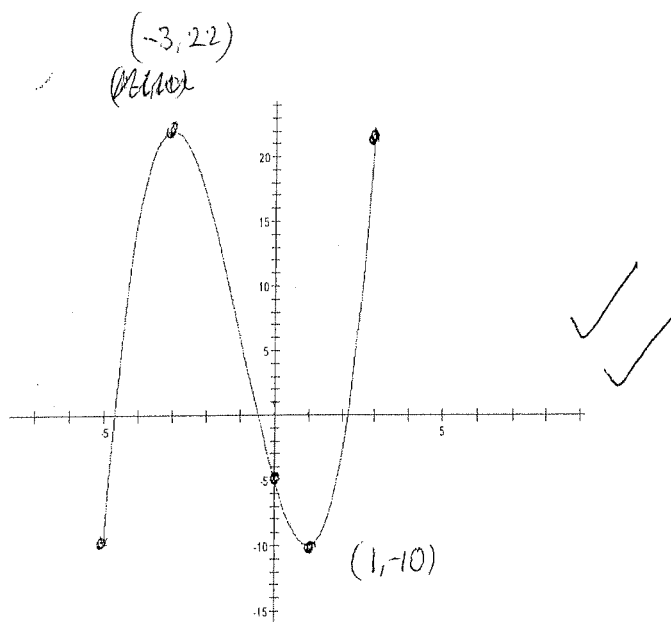
At $x = -3$ $\frac{d^2y}{dx^2} = -12 < 0$

$\Rightarrow \text{Max}$ ✓

At $x = 1$ $\frac{d^2y}{dx^2} = 12 > 0$

$\Rightarrow \text{Min}$ ✓

iv.



6.

$$y' = -3x^2 e^{-x^3}$$
 ✓

$$\Rightarrow \int_1^2 x^2 e^{-x^3} dx = \left[\frac{e^{-x^3}}{-3} \right]_1^2 = \left[\frac{(e^{-8} - e^{-1})}{-3} \right] = 0.123$$

b. $\int_0^1 e^{2-3x} dx = \left[\frac{e^{2-3x}}{-3} \right]_0^1 = \frac{e^{-1} - e^2}{-3} = \frac{e^3 - 1}{3e}$

Question 8

Let P: (x, y), then:

$$PA = \sqrt{(x+6)^2 + (y+2)^2}$$

$$PB = \sqrt{(x-2)^2 + (y-2)^2}$$

Since $PA=3PB$

$$\sqrt{(x+6)^2 + (y+2)^2} = 3\sqrt{(x-2)^2 + (y-2)^2} \quad /(\)^2 \quad \checkmark (1)$$

$$(x+6)^2 + (y+2)^2 = 9((x-2)^2 + (y-2)^2)$$

$$x^2 + 12x + 36 + y^2 + 4y + 4 = 9(x^2 - 4x + 4 + y^2 - 4y + 4) \quad \checkmark (1)$$

$$x^2 + 12x + y^2 + 4y + 40 = 9(x^2 - 4x + y^2 - 4y + 8)$$

$$x^2 + 12x + y^2 + 4y + 40 = 9x^2 - 36x + 9y^2 - 36y + 72$$

$$0 = 8x^2 - 48x + 8y^2 - 40y + 32 \quad \checkmark (1)$$

$$0 = x^2 - 6x + y^2 - 5y + 4$$

$$0 = (x-3)^2 - 9 + (y-2.5)^2 - 6.25 + 4 \quad \checkmark (1)$$

$$11.25 = (x-3)^2 + (y-2.5)^2$$

Equation of circle with centre (3, 2.5) and radius $\frac{3\sqrt{5}}{2}$ $\checkmark (1)$

b. Rearrange: $x = \frac{3}{y}$

Volume generated is given by:

$$V = \pi \int_1^4 \left(\frac{3}{y}\right)^2 dy \quad \checkmark (1)$$

$$V = \pi \int_1^4 \left(\frac{9}{y^2}\right) dy \quad \checkmark (1)$$

$$V = \pi \int_1^4 (9y^{-2}) dy$$

$$V = \pi \left[\frac{9y^{-1}}{-1} \right]_1^4 \quad \checkmark (1)$$

$$V = \pi \left[-\frac{9}{y} \right]_1^4 = \pi \left(\frac{-9}{4} + 9 \right) = 21.206u^3 \quad \checkmark (1)$$

c.
d.

Question 9

a.

i. $y' = 4(e^{4x+3} - 3)^3 \times 4e^{4x+3}$

$y' = 16e^{4x+3}(e^{4x+3} - 3)^3$

ii. Rearrange: $y = xe^{-x}$ Use product rule with:

$u = x \quad v = e^{-x}$

$u' = 1 \quad v' = -e^{-x}$

$y' = 1e^{-x} - xe^{-x}$

$y' = e^{-x}(1 - x)$

b. For A and C need to solve: $(x+1)(x-1)^2 = 0$

$x = -1$ or $x = 1$

$A: (-1, 0) \quad B: (1, 0)$

For B , substitute $x=0$ in equation: $y = (0+1)(0-1)^2 = 1$

$B: (0, 1)$

c. Rewrite: $y = x^3 - x^2 - x + 1$

$A_1 = \int_{-1}^0 (x^3 - x^2 - x + 1) dx$

$A_1 = \left[\frac{x^4}{4} - \frac{x^3}{3} - \frac{x^2}{2} + x \right]_{-1}^0$

$A_1 = \frac{11}{12}$

$A_2 = \int_0^1 (x^3 - x^2 - x + 1) dx$

$A_2 = \left[\frac{x^4}{4} - \frac{x^3}{3} - \frac{x^2}{2} + x \right]_0^1$

$A_2 = \frac{5}{12}$

$\frac{A_1}{A_2} = \frac{\frac{11}{12}}{\frac{5}{12}} = \frac{11}{5}$

b. $\int_1^5 x \log_e x \approx \frac{1}{3} (0 \times 1 + 1.39 \times 4 + 3.30 \times 2 + 5.55 \times 4 + 8.05 \times 1)$

$\int_1^5 x \log_e x \approx 14.14$

c.

Question 10

a.

i. Consider: $f(x) = \frac{1}{2}(e^x + e^{-x})$

Then: $f(-x) = \frac{1}{2}(e^{-x} + e^x)$

$\therefore f(-x) = f(x)$ The function is even.

ii. $f'(x) = \frac{1}{2}(e^x - e^{-x})$

For stationary point $f'(x) = 0 \Rightarrow 0 = \frac{1}{2}(e^x - e^{-x})$

$\Rightarrow 0 = e^x - e^{-x}$

$\Rightarrow e^x = e^{-x}$

$\Rightarrow x = -x$

$\Rightarrow 2x = 0$

$\Rightarrow x = 0$

At $x=0$: $f(0) = \frac{1}{2}(1+1) = 1$

$\Rightarrow (0,1)$ stationary point

Nature of stationary point.

$f''(x) = \frac{1}{2}(e^x + e^{-x})$

At $x=0$

$f''(0) = 1 > 0 \Rightarrow \text{Minimum}$

iii. For inflexion, need to solve:

$f''(x) = 0$

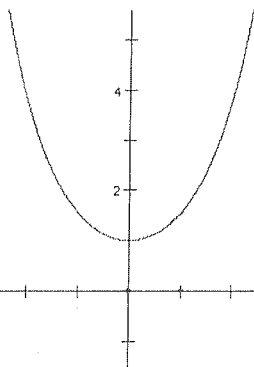
$0 = \frac{1}{2}(e^x + e^{-x})$

$0 = e^x + e^{-x}$

No solutions since both

$e^x > 0$ and $e^{-x} > 0$ for all values of x

Hence no points of inflexion.



iv.



b.

i. Substitute $x=2$ into the function:

$y = e^{-4} + e^2 = 7.41$

ii. $A = \int_0^2 (e^{2x} + e^x) dx$

$A = \left[\frac{e^{2x}}{2} + e^x \right]_0^2 = 6.880u^2$